

Learning How to Learn with Prof. Barbara Oakley

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ABSTRACT

Massive Online Open Courses or MOOCs are increasingly becoming visible with the advent of platforms like Coursera, which offer verified certificates. One of the more popular courses on Coursera is 'Learning How to Learn', which specifically targets students and provides them with a scientific basis for effective pedagogical techniques. The course is the brainchild of Prof. Barbara Oakley and her co-instructor, Terrence Sejnowski, the Francis Crick Professor at the Salk Institute. I recently had the opportunity to interview Prof. Barbara Oakley. This article is a summary of her responses.

Categories and Subject Descriptors

K.3.1 [Computers and Education]: Computer Uses in Education – Collaborative Learning

General Terms

Human Factors

Keywords

Education, MOOC, WIC.

1. INTRODUCTION

Learning How to Learn (LHL) is a Coursera offering through the University of California, San Diego. Students can start LHL any time they wish, and proceed at their own pace. They can even get a certificate to establish that they have completed the requirements for the course with a passing grade (~80%). Since LHL was first offered, it has become the most popular offering from Coursera. The course was developed by Prof. Barbara Oakley and her co-instructor, Terrence Sejnowski, the Francis Crick Professor at the Salk Institute. Dr. Oakley's work on the course has been recognized via numerous awards, including Coursera's Inaugural Innovation Instructor, the American Society of Engineering Education Chester F. Carlson Award for innovation in STEM education, and the Theo C. Pilkington Outstanding Educator Award for Pedagogical Advances in Bioengineering Education. I recently had the opportunity to speak with Prof. Barbara Oakley and ask her a few questions. This article is a summary of her responses.

2. BACKGROUND

Prof. Oakley joined the army right out of high school, where she learned Russian at the Defense Language Institute. She won an in-service scholarship that eventually led to her being commissioned in the US Army Signal Corps. She had done poorly in math and science in high school, but one of her assignments required her to work with West Point engineers, who impressed her with their problem solving skill set. So at the age of 26, Prof. Oakley began the preliminary training she needed to enroll in an undergraduate engineering program. During her studies she realized that the

language-learning approaches she'd acquired while learning Russian could also be useful to help her to learn math and science. Thus, despite her limited background in math, she managed to excel. The relationship between learning in verbal and mathematical domains later became the subject of her *New York Times* best-selling science book, *A Mind for Numbers: How to Excel at Math and Science*.

3. CURRENT RESEARCH

During her army days, Prof. Oakley spent substantial amount of time working with Soviet nationals. Given the harsh conditions under the former U.S.S.R., she began to wonder why "nasty people do what they do." Thus, while pursuing tenure as a professor of engineering, she started investigating the neuroscientific and psychological research related to this question. This became the subject of her tongue-in-cheek titled, but critically-praised book *Evil Genes: Why Rome Fell, Hitler Rose, Enron Failed, and My Sister Stole My Mother's Boyfriend*.

In the process of writing the book, Dr. Oakley was startled to find that many established ideas in social science have no theoretical grounding. One of these ideas is the implicit assumption that altruism is inherently and always good. Her experience with the Soviets and with communism led her to suspect that good intentions could, in fact, lead to substantive harm. Thus, her research switched to asking the question, "why do nice people follow or align with nasty people, and what harm can arise from good intentions, or perceived good intentions?" This became the subject of her next two books: *Cold Blooded Kindness* (Prometheus Books, 2011) and *Pathological Altruism* (Oxford University Press, 2012).

4. INSPIRATION FOR LHL

While volunteering with students in a local inner urban school district, Prof. Oakley realized that many of the pedagogical approaches being used were deeply flawed. She also recognized that trying to change the system itself would be a time and resource intensive task. Her alternative was to eventually, with Dr. Sejnowski, create a Massive Open Online Course (MOOC); a course that can be accessed, not just by the students of a specific inner school district, but also by students across the world.

She knew that most online courses related to education were typically geared towards educators rather than students—yet students were the ones who needed to improve their learning skills. So LHL was targeted directly towards students, in particular to help provide them with tips that could help build a passion for learning based on self-efficacy and internal motivation.

5. BENEFITS AND LIMITATIONS OF MOOCs

MOOCs allow students from one part of the world, e.g. Japan, to interact with peers across the globe, e.g. Venezuela. In addition to

geographic boundaries, MOOCs also break down boundaries across other demographic variables, such as age. Thus, MOOCs are typically more diverse than face-to-face educational institutions.

MOOCs also reduce the monopoly of teachers, by providing students an alternative to bad teachers. Being stuck with poor instructors can limit a student's potential, to the extent that the student no longer feels that they can have even a basic grasp of the subject. MOOCs provide motivated students an alternative that can help them to learn and excel.

However, MOOCs are currently limited by lack of ability of students to receive college credit for taking them. Enabling students to receive college credits for a MOOC through a university can be a herculean task. This is partly due to education policies that haven't caught up to advances in technology, partly due to limitations in the current technology, and partly due to antipathy by professors towards a medium that might reduce the footprint of the professorate on society. But changes in this area are happening swiftly.

6. CHALLENGES FOR WOMEN IN COMPUTING

The challenges for women in computing and technical fields have evolved significantly since Prof. Oakley was a student. She states, "Women and men have basically the same abilities in math and science. The difference is that testosterone in men can reduce men's verbal abilities, while the *lack* of testosterone in women enhances women's verbal abilities. This means that women *feel* as if they're better at verbal activities than they are at math and science, and men *feel* as if they're better at math and science than they are in verbal areas—even though both groups have basically the same abilities in math and science!" Dr. Oakley goes on to note: "The issue is that we tend to tell women to follow their passions for what they feel good at, while we should really be telling them to *broaden* their passions—to keep going in math and science even though sometimes verbal areas may feel easier for them."

7. AUTHOR'S PERSONAL NOTE

My personal experience with the Learning How to Learn course has been both personally and professionally rewarding. I was amazed both by the breadth of topics discussed as well as the intuitive nature of the various analogies used. The lessons I learned from the course have led to substantial changes in how I approach my day. One of the lessons imparted in the course was that we should try to tackle our most difficult task for the day the first thing in the morning, leaving easier tasks for later. As a result, I no longer reply to emails as soon as reach the office, which usually tends to be the easiest task. Instead I leave email replies for post lunch, when my brain is still recovering from a food induced coma.

Many of the lessons in the course will be intuitive and perhaps even familiar to the people reading this article. LHL provides the scientific foundations behind these. In addition it introduces us to all the tips and tricks to learning that we didn't know. I believe that LHL is one of the few courses that all of us should take, not the least because it provides a foundation for learning that will allow us to learn other new things. Hopefully, this article will motivate you to test my claims yourself. If so, check out www.coursera.org/course/learning.

8. ACKNOWLEDGEMENT

I would like to thank Prof. Barbara Oakley for taking the time to talk to me. To learn more about Prof. Oakley's work, please check out her books – [A Mind for Numbers](#), [Evil Genes](#), [Cold-Blooded Kindness](#), and [Pathological Altruism](#).